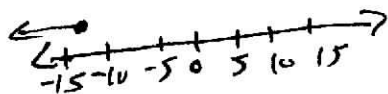


Unit 6: Linear Inequalities

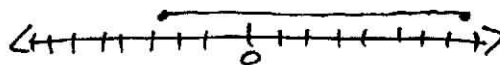
1. Solve and graph the inequality: $\frac{2}{3}(12m + 108) \leq 4 - (-4m - 12)$



$$\begin{aligned} 8m + 72 &\leq 4m + 16 \\ -4m &\quad -4m \\ \hline 4m + 72 &\leq 16 \\ -72 &\quad -72 \\ \hline 4m &\leq -56 \\ \frac{4m}{4} &\leq \frac{-56}{4} \\ m &\leq -14 \end{aligned}$$

2. Solve and graph the compound inequality: $-\frac{4}{5} \leq \frac{2x-1}{3} + \frac{3}{2} \leq 6$

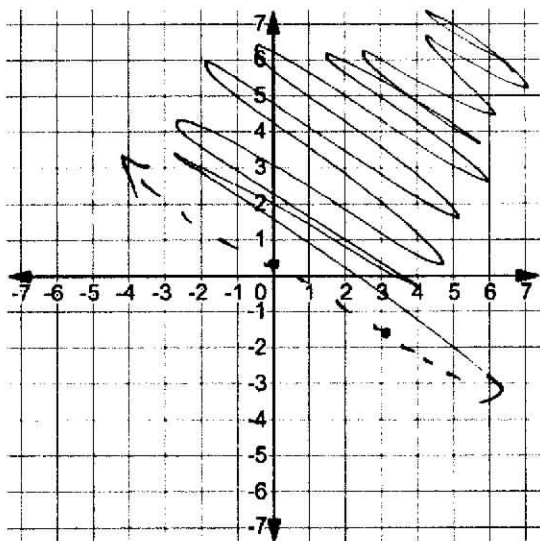
$$\begin{aligned} -\frac{4}{5} &\leq \frac{2x}{3} + \frac{7}{6} \leq 6 \\ -\frac{7}{6} &\quad -\frac{7}{6} \quad -\frac{7}{6} \\ \hline -\frac{59}{30} &\leq \frac{2x}{3} \leq \frac{29}{6} \\ \frac{-59}{30} \cdot \frac{3}{2} &\quad \frac{29}{6} \cdot \frac{3}{2} \\ \frac{-59}{20} &\leq x \leq \frac{29}{4} \\ -2.95 &\leq x \leq 7.25 \end{aligned}$$



3. In the spring a farmer lets you pick strawberries for \$2 a pound and blueberries for \$3 per pound. Aaron has \$25 to spend. Create a linear inequality to represent this situation.

$$2s + 3b \leq 25$$

4. Graph the inequality $2x + 3y > 2$



$$\begin{aligned} 2x + 3y &> 2 \\ -2x &\quad -2x \\ \hline 3y &> -2x + 2 \\ \frac{3y}{3} &\quad \frac{-2x}{3} \quad \frac{2}{3} \\ \hline y &> -\frac{2}{3}x + \frac{2}{3} \end{aligned}$$